UMJI FA22 VG101 Homework 6

Assigned: 8 Nov. 2022 Due: 15 Nov. 2022 23:59pm

Notes for Submission

Please submit to the JOJ for grading purpose (**follow the naming rule on JOJ**).

For backup purpose, you should also submit your homework to the Canvas. For canvas submission, please name the file using the format: `hw6-[Your Last Name]-[Your First Name]-[Your SJTU ID].c` For example, `hw6-Yang-Zijun-521370910109.c`

To make your life easier while working on the exciting lab problems, this homework problem will be easy. Nevertheless, if you are interested and have additional time, you may try to use recursion, which is not required, to implement it.

Homework Problem

Read the following program and complete the function `sortstr` to sort the numbers in ascending order in a character string str, which is an array of positive integers separated by spaces. To make your life easier, we assume that there is only one space between the numbers, and there is no space before the first number or after the last number. The output str should have similar format. In addition, we assume that at least 1 number and at most 10 numbers will be input and the total number of characters in the input string is less than 255. For example, if the user input is `3 5 2 7 4`, then the output should be `2 3 4 5 7`. You don't need to consider invalid inputs. Hint: to assemble the output str, you may consider using the function `sprintf` and `strcat`.

```c
// You can include libraries you need
#include <stdio.h>
#include <string.h>


// hw6 - finish this function
void sortstr(char str[]) {
    // TODO: write your code here

}
```

```c
// Do NOT change the following lines.
int main() {
    char str[256];
    scanf("%[^\n]%*c", str);  // ask for the user input
    sortstr(str);
    printf("%s\n", str);      // output
    return 0;
}
```

The following is an example run of the program:

```
3 5 2 7 4
2 3 4 5 7
```

where `3 5 2 7 4` is the user input and `2 3 4 5 7` is the printed output.